

Lesson 6-3 · Single-Period Quick

Logarithms + DOK-3 Spine: q15 ($\exp \leftrightarrow \log$ inverse via graph)

Klimsara-adapted · Student-centered · No Blooket

DOK 1–2

7 min

bridge to logarithms launch

Three items — work solo

1. **q14 Error Analysis:** Find and correct the error in this work:
 $16e^t = 98 \rightarrow e^t = 6.125 \rightarrow 6.125t = \ln e$
2. **q16 Generalize:** For what values of x is $\log_4 x < 0$ true?
3. **q3 Vocabulary:** Explain the difference between the common log and the natural log.

Launch — Example 3: Evaluate Logarithms

Algebra 2 · Lesson 6-3

DOK 1–2

8 min

teacher think-aloud Example 3 (Evaluate Logarithms)

LOG $\leftrightarrow$ EXP RULES (keep printed on your page) $\log_b x = y \Leftrightarrow b^y = x$

COMMON LOG: $\log x = \log_{10} x$

NATURAL LOG: $\ln x = \log_e x$

KEY: $\ln(e^t) = t$ UNDEFINED: $\log_b 0$ and $\log_b(\text{negative})$ IDENTITY: $\log_b(b^k) = k$

Teacher think-aloud — four cases

$$\log_5 125 = 3 \quad (5^3 = 125)$$

$$\log_{1/4} 16 = -2 \quad ((1/4)^{-2} = 4^2 = 16; \text{flag: negative})$$

$$\log_3 0: \text{undefined} \quad (\text{no power of 3 reaches 0})$$

$$\log_2 2^8 = 8 \quad (\text{identity: } \log_b(b^k) = k)$$

30 sec pause: Rewrite $\log_2 64$ using the definition with your partner

Explore — TPS + DOK-3 Spine (q15)

Algebra 2 · Lesson 6-3

DOK 2–3

25 min

TPS + DOK-3 driver

5 items in order. Plan ~10 min for q15 (DOK-3) + 5 min pair presentations.

Practice #15 · DOK-3 SPINE (★) Use the graph of $y = 3^x$ ($x \in [-1, 4.5]$, $y \in [0, 85]$) to **estimate** $\log_3 50$. Explain your reasoning. *(graph on student packet)*

Practice #52 · Continuous growth (q52) How long for \$250 to grow to \$600 at 4% APR compounded continuously? Round to nearest year.

Practice #53 · Compare accounts (q53) Michael vs. Peter: who reaches \$1,800 first? (45-min variant: skip)

DOK 2

5 min

academic conversation + 6-4 bridge

Sentence frame · q15 pair-share “On the graph of $y = 3^x$, I found $\log_3 50$ by looking for $y = \underline{\hspace{1cm}}$ and reading the x-value. Since $3^3 = \underline{\hspace{1cm}}$ and $3^4 = \underline{\hspace{1cm}}$, the answer is between $\underline{\hspace{1cm}}$ and $\underline{\hspace{1cm}}$, closer to $\underline{\hspace{1cm}}$. This works because $\log_3 50 = x$ means $3^x = \underline{\hspace{1cm}}$, so logs and exponentials are _____.”

Bridge to Lesson 6-4 Next lesson: logarithmic **FUNCTIONS** — graphing $y = \log_b x$, domain, range, asymptotes. Today's log $\leftrightarrow$ exp definition is the foundation.

DOK 1–2

5 min

summary recap (not graded)

Complete on your exit ticket

1. $\log_b x = y$ means _____.
2. $\log_2 32 = \underline{\hspace{1cm}}$ because $2^? = 32$.
3. To solve $e^t = 6.125$, take the _____ of both sides giving $t = \underline{\hspace{1cm}}$.
4. One biggest thing I learned today: _____